

Bayesian Hierarchical Model for PM_{2.5} levels in Italy

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1 Introduction

Air pollution represents a major environmental and public health concern. Specifically, fine particulate matter $\text{PM}_{2.5}$ is among the most harmful pollutants due to its ability to penetrate deep into the respiratory system and its strong association with cardiovascular and pulmonary diseases.

A key challenge in modeling $\text{PM}_{2.5}$ concentrations lies in the presence of substantial heterogeneity across monitoring stations and irregular temporal coverage. Stations differ significantly in terms of geographical characteristics, local emission sources, and environmental conditions, while observational data often exhibit an unbalanced panel structure, with varying numbers of observations across stations and missing time periods.

In this context, Bayesian hierarchical models provide a natural and flexible framework to account for these complexities. By introducing group-level random effects, these models allow for partial pooling, enabling information sharing across stations while preserving local heterogeneity. Moreover, the Bayesian approach provides a coherent framework for uncertainty quantification and prior specification.

This paper develops a Bayesian hierarchical model for daily $\text{PM}_{2.5}$ concentrations observed across monitoring stations in Italy. Posterior inference is carried out using a Metropolis-within-Gibbs sampler. The analysis focuses on model specification, MCMC implementation, convergence diagnostics, sensitivity analysis, and posterior predictive checks.

2 Data

The dataset comprises daily $\text{PM}_{2.5}$ concentration measurements collected across monitoring stations in Italy during 2025. The raw data were sourced from the European Environment Agency (EEA) Air Quality portal, specifically via the EEA Air Quality e-Reporting, with metadata and station specifications validated through the EEA Air Quality Viewer. After preprocessing and quality control, the final dataset contains 31,349 observations distributed across 116 stations.

The empirical distribution of the raw outcome variable is highly right-skewed, with a skewness of approximately 17.1. This strong asymmetry is primarily driven by extreme pollution events in the upper tail. To stabilize the variance and satisfy the distributional assumptions of the proposed model, a logarithmic transformation was applied

$$\log(\text{PM}_{2.5,it}) = y_{it}$$

where y_{it} represents the transformed concentration for station i at time t . This transformation substantially reduces skewness (to approximately -0.29), yielding a more symmetric distribution (visible in Figure 1) that justifies the use of a Gaussian likelihood.

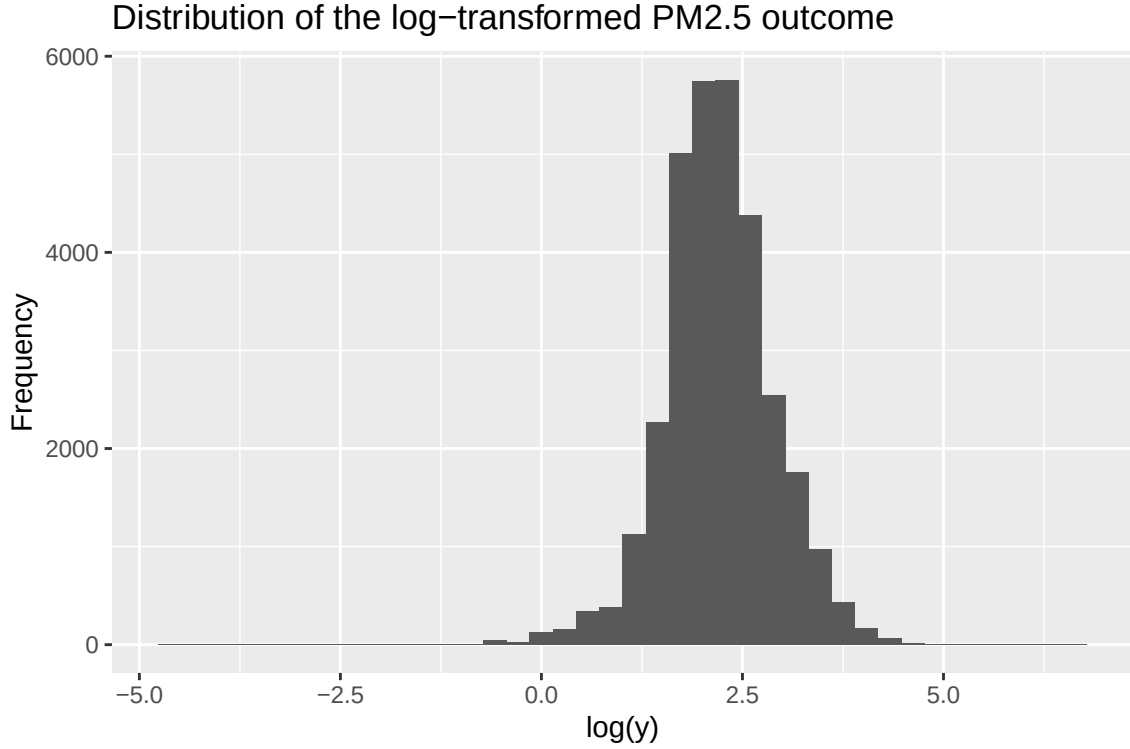


Figure 1: Distribution of raw and log-transformed PM2.5 concentrations.

The panel structure is markedly unbalanced, a common feature in environmental monitoring networks. The number of observations per station ranges from 69 to 332, with a median of approximately 299. Temporal coverage is also uneven across stations. While most stations provide observations for most of the year, 32 stations lack at least one full month of data, and for a few stations coverage drops to only 4 months. This heterogeneity is summarized in Figure 2, which reports both the distribution of the number of observations per station and the distribution of the number of months observed per station. Such an irregular structure poses a challenge for standard time-series methods.

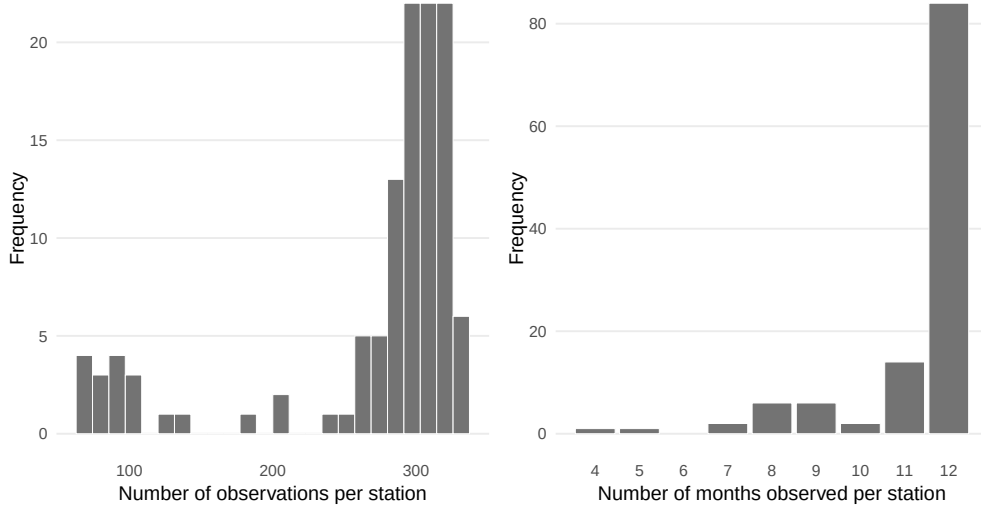


Figure 2: Data availability by station and month. Darker areas indicate months with fewer observations.

The hierarchical nature of the data and the irregular sampling motivate the use of a Bayesian hierarchical specification, since such a framework is naturally suited to handle unbalanced grouped data by leveraging information across the entire network while accounting for station-specific nuances.

3 Model

We consider a hierarchical model for the log-transformed PM2.5 concentrations. Let $\tilde{y}_i$ denote the transformed outcome for observation i . The likelihood is specified as:

$$\tilde{y}_i \mid \beta, \alpha, \sigma^2 \sim \mathcal{N}(x_i^\top \beta + \alpha_{j[i]}, \sigma^2),$$

where: - x_i is a vector of covariates including month and day-of-week effects, - β is a vector of regression coefficients, - $\alpha_{j[i]}$ is a station-specific random intercept.

3.1 Hierarchical structure

The random effects are modeled as:

$$\alpha_j \mid \tau \sim \mathcal{N}(0, \tau^2), \quad j = 1, \dots, J,$$

which induces partial pooling across stations.

3.2 Prior specification

Weakly informative priors are assigned to all parameters.

For the regression coefficients:

$$\beta \sim \mathcal{N}(b_0, B_0),$$

where $B_0 = cI$ is a diagonal covariance matrix.

For the observation variance:

$$\sigma^2 \sim \text{Inverse-Gamma}(a_\sigma, b_\sigma).$$

For the group-level standard deviation:

$$\tau \sim \text{Half-Cauchy}(0, A).$$

The use of a half-Cauchy prior for τ avoids the well-known sensitivity of inverse-gamma priors for variance components, while providing weakly informative regularization.

3.3 Posterior inference

Posterior inference is performed using a Metropolis-within-Gibbs sampler. The full conditional distributions for β , α_j , and σ^2 belong to standard families and are sampled via Gibbs steps.

In contrast, the full conditional distribution of τ is not available in closed form due to the half-Cauchy prior. Therefore, a Metropolis-Hastings step is used, implemented on the logarithmic scale with a normal random-walk proposal.

The sampling scheme can be summarized as follows:

- Sample β from its conditional posterior
- Sample each α_j from its conditional posterior
- Sample σ^2 from its conditional posterior
- Update τ via a Metropolis-Hastings step

4 MCMC Implementation

The sampler is run for 5,000 iterations. Visual inspection of the trace plots suggests a short initial transient phase, and the first 200 iterations are discarded as burn-in. To reduce autocorrelation, a thinning scheme is applied by retaining every 5th draw.

4.1 Tuning of the Metropolis step

The proposal distribution for τ is a normal random walk on the log scale. A tuning exercise was conducted to assess the impact of the proposal standard deviation.

With a proposal standard deviation of 0.15, the acceptance rate was approximately 0.48. Increasing the proposal scale to 0.20 reduced the acceptance rate to approximately 0.38. Despite this change, trace plots and autocorrelation functions remained essentially unchanged, and posterior summaries were virtually identical.

Given these results, the value `proposal_sd = 0.20` is retained in the final implementation.

5 Results

5.1 Convergence diagnostics

Trace plots of the post-burn-in chains are shown in Figure 3. There is no evidence of non-stationarity or systematic trends. The chains fluctuate around stable levels, indicating that convergence has been achieved.

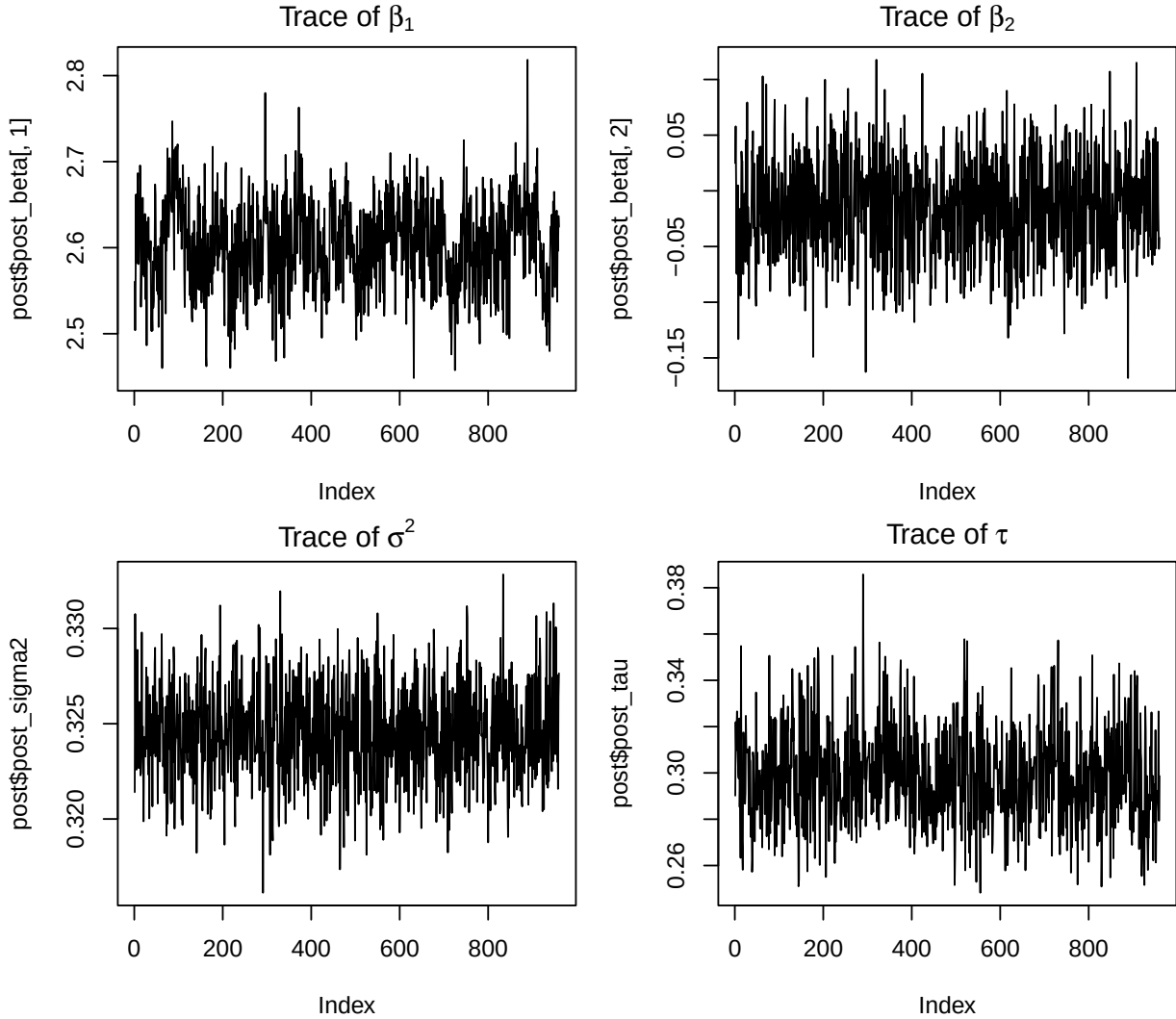


Figure 3: Post-burn-in trace plots for selected model parameters. The chains fluctuate around stable levels without visible trends, supporting convergence of the Metropolis-within-Gibbs sampler.

Residual autocorrelation is illustrated in Figure 4, motivating the use of thinning in posterior summaries.

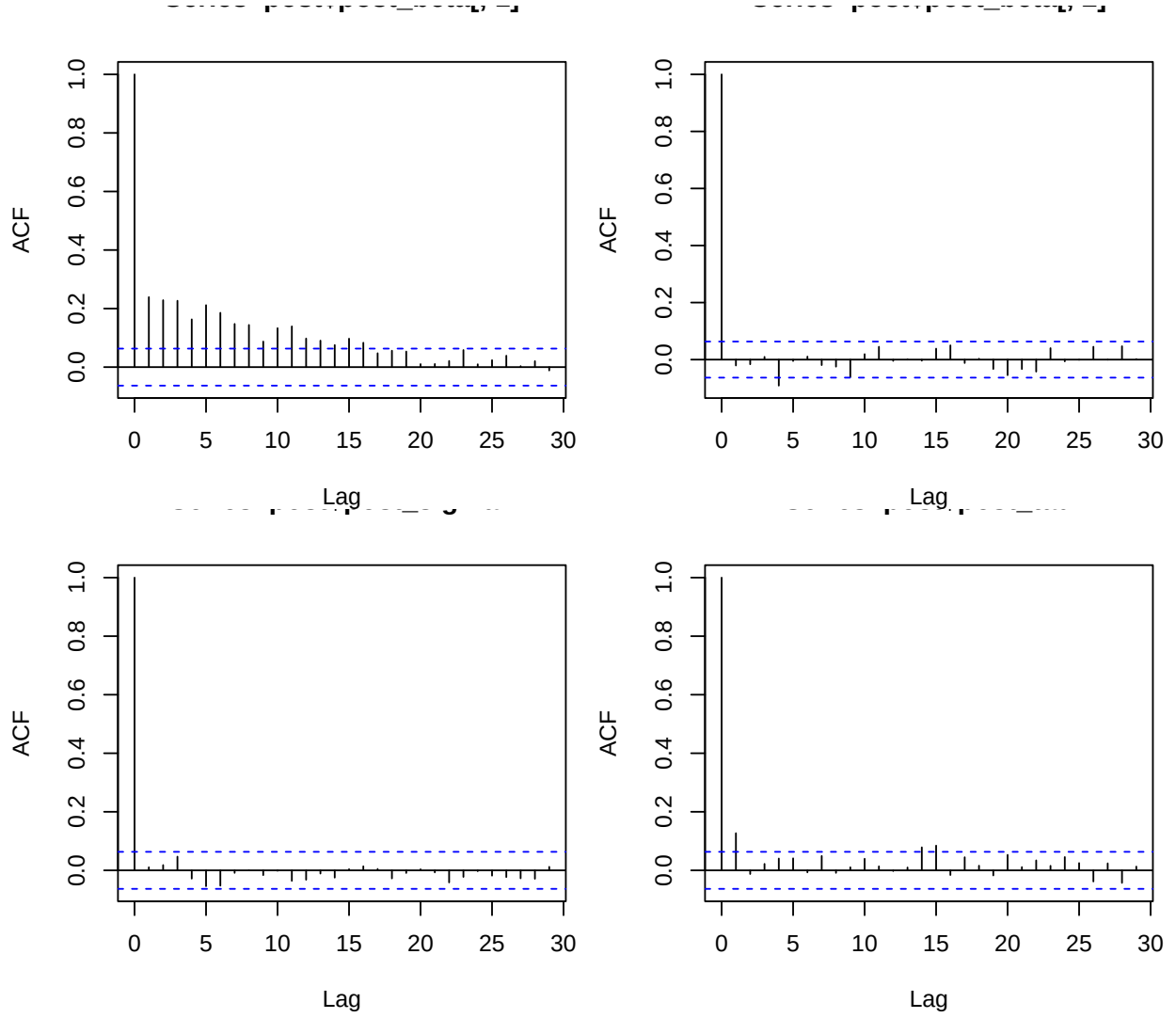


Figure 4: Post-burn-in autocorrelation functions for selected model parameters. Residual autocorrelation is limited for most parameters, although some mild dependence persists.

The Geweke diagnostic confirms this result. All z-scores are well within the conventional acceptance range, providing no evidence against convergence.

The effective sample size (ESS) is generally high across parameters. In particular, the variance parameter exhibits an ESS close to the total number of retained draws, while the hyperparameter τ also shows a high ESS. The random effects display more variability, as expected, but remain within acceptable ranges.

5.2 Posterior summaries

The posterior estimates shown in Table 1 provide a clear and economically meaningful interpretation of the main drivers of PM2.5 concentrations.

The intercept represents the expected value of the log-transformed outcome for the reference category, corresponding to January 2025 and Monday, for a station with average random effect. Its posterior mean is approximately 2.60, which translates into a baseline PM2.5 level of about 13.5 on the original scale. This value can be interpreted as a typical baseline level of air pollution under winter conditions.

Seasonal effects.

A strong and systematic seasonal pattern emerges from the estimated monthly coefficients. January serves as the reference month, and nearly all other months exhibit negative coefficients relative to it. This indicates that PM2.5 concentrations tend to be significantly lower throughout most of the year compared to January.

The magnitude of these effects is substantial. In particular, spring, summer, and early autumn months are associated with large reductions in PM2.5 levels, often exceeding 30–50% relative to January. December, on the other hand, appears statistically close to January, suggesting a return to high pollution levels during the winter season.

This pattern is consistent with well-known environmental mechanisms. During colder months, increased emissions from residential heating, combined with more stable atmospheric conditions and reduced dispersion, lead to the accumulation of particulate matter. In contrast, warmer months are characterized by improved atmospheric mixing and lower heating-related emissions.

Day-of-week effects.

The estimated coefficients for the day of the week are small in magnitude and their credible intervals include zero. This suggests that, once seasonal effects and station-level heterogeneity are accounted for, there is no strong systematic weekly cycle in PM2.5 concentrations.

Variance decomposition and spatial heterogeneity.

The posterior estimate of the residual standard deviation is larger than the group-level standard deviation, indicating that day-to-day fluctuations within a station are more pronounced than persistent differences across stations. However, the between-station variability remains non-negligible, highlighting structural geographical heterogeneity.

Economic and geographical implications.

The results suggest that pollution exposure is strongly seasonal, with higher risk concentrated in winter months. This supports the use of seasonally targeted environmental policies, particularly focusing on emissions from heating and transport.

At the same time, the absence of strong weekday effects indicates that policies based solely on

Table 1: Posterior means and approximate percentage effects relative to the baseline (January, Monday).

Variable	Mean	Effect
Intercept (Jan, Mon)	2.603	Baseline (~13.5)
February	-0.016	-1.6%
March	-0.362	-30.4%
April	-0.615	-45.9%
May	-0.742	-52.4%
June	-0.119	-11.3%
July	-0.598	-45.0%
August	-0.425	-34.6%
September	-0.799	-55.0%
October	-0.605	-45.4%
November	-0.381	-31.7%
December	0.068	+7.0%

weekly cycles may have limited impact unless combined with broader structural interventions.

Finally, the presence of persistent differences across stations highlights the importance of geographically differentiated policies.

5.2.1 Spatial Analysis

Figure 5 displays the geographical distribution of monitoring stations, with colors representing the posterior mean of the station-specific random effects.

The spatial comparison is based on a simple latitude-based grouping and should be interpreted as an exploratory geographical characterization rather than a formal spatial model. Each point in the map represents a station, with color indicating the posterior mean of the station-specific random effect. Positive values correspond to locations with structurally higher PM2.5 levels after controlling for temporal effects, while negative values indicate lower baseline pollution levels.

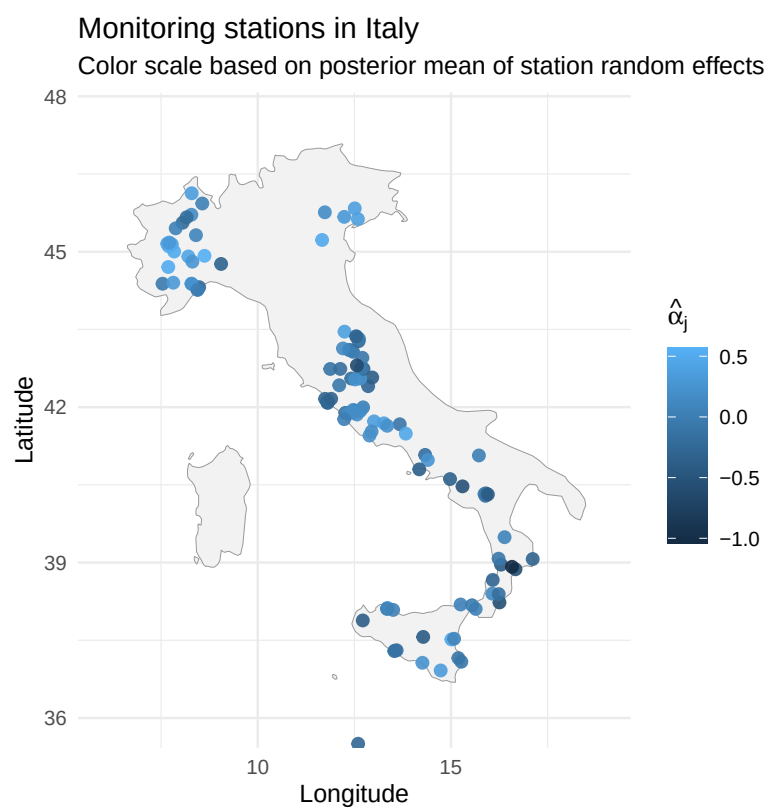


Figure 5: Map of monitoring stations in Italy.

A coarse geographical comparison based on station latitude reveals a clear North–South gradient in the posterior station effects. Stations located in the northern part of Italy display, on average, positive random effects, whereas southern stations tend to exhibit negative values, with central stations lying in between.

A posterior comparison between the average station effects in the North and in the South shows a mean difference of about 0.31 on the log scale, with the corresponding 95% credible interval entirely above zero. The posterior probability that the North has a higher average station effect than the South is effectively equal to one. On the original scale, this difference corresponds to roughly a 36% higher structural PM2.5 level in the North relative to the South, after controlling for month and day-of-week effects.

This finding is consistent with the idea that northern Italy is more exposed to persistent pollution accumulation mechanisms, likely due to a combination of higher urban and industrial intensity, traffic exposure, and less favorable dispersion conditions. At the same time, the spatial distribution of the station effects also shows substantial local heterogeneity, suggesting that broad geographical gradients coexist with important location-specific determinants of air quality.

5.3 Sensitivity analysis

A sensitivity analysis was conducted by varying key prior hyperparameters, including: - the prior variance of β , - the scale parameter of the half-Cauchy prior on τ , - the parameters of the inverse-gamma prior on σ^2 .

The results show that posterior estimates are remarkably stable across all specifications. In particular, the posterior means of the intercept, σ^2 , and τ remain virtually unchanged. This suggests that the data are highly informative and that the posterior distribution is largely driven by the likelihood.

5.4 Posterior predictive checks

Posterior predictive checks were performed to assess model adequacy. Replicated datasets were generated from the posterior distribution and compared to the observed data.

The distribution of replicated means and standard deviations is consistent with the observed values (see Figures @ref(fig:fig-ppc-density and 7, indicating that the model accurately captures the first two moments of the data.

Furthermore, the posterior predictive density aligns well with the empirical distribution of log-transformed PM2.5, with only minor discrepancies in the peak and tail regions. These differences are consistent with the Gaussian assumption and do not indicate major model misspecification.

6 Conclusion

This paper develops a Bayesian hierarchical model for daily PM2.5 concentrations, accounting for heterogeneity across monitoring stations and unbalanced panel structure.

The Metropolis-within-Gibbs sampler exhibits good convergence properties, supported by multiple diagnostic tools. Posterior inference is robust to prior specification, and posterior predictive checks indicate a satisfactory model fit.

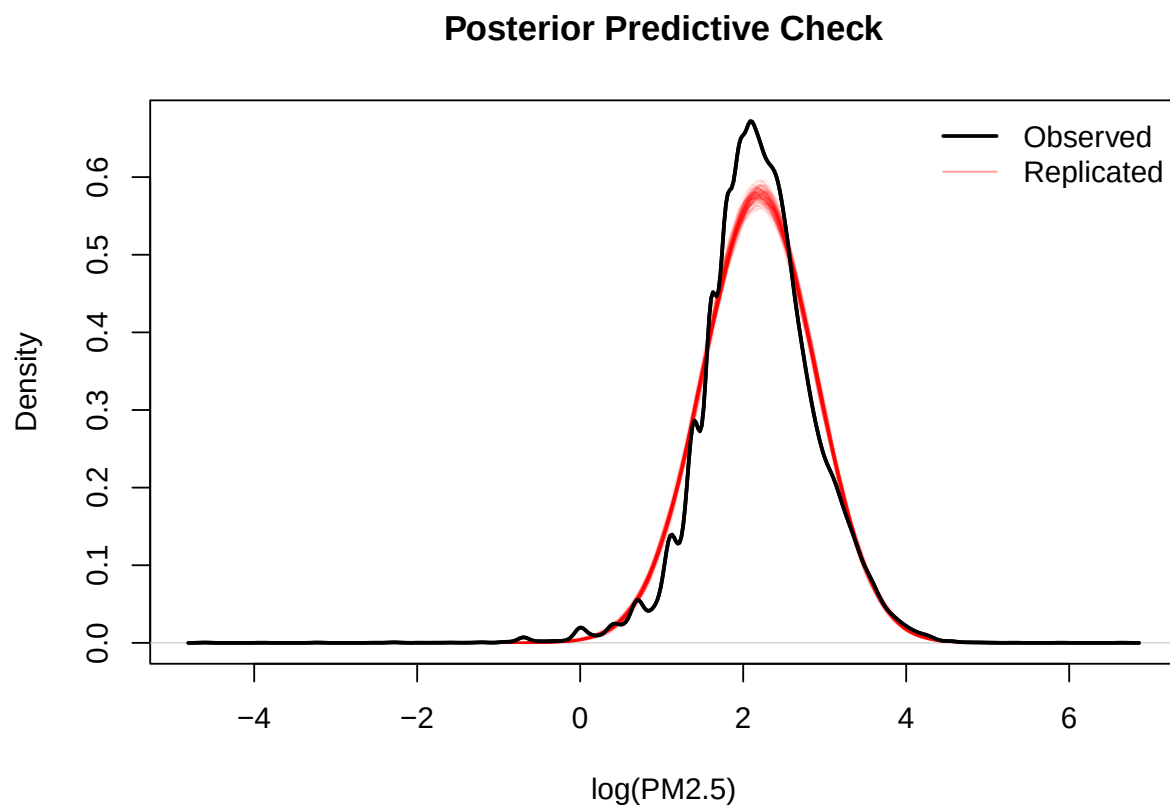


Figure 6: Posterior predictive check for the log-transformed PM2.5 outcome.

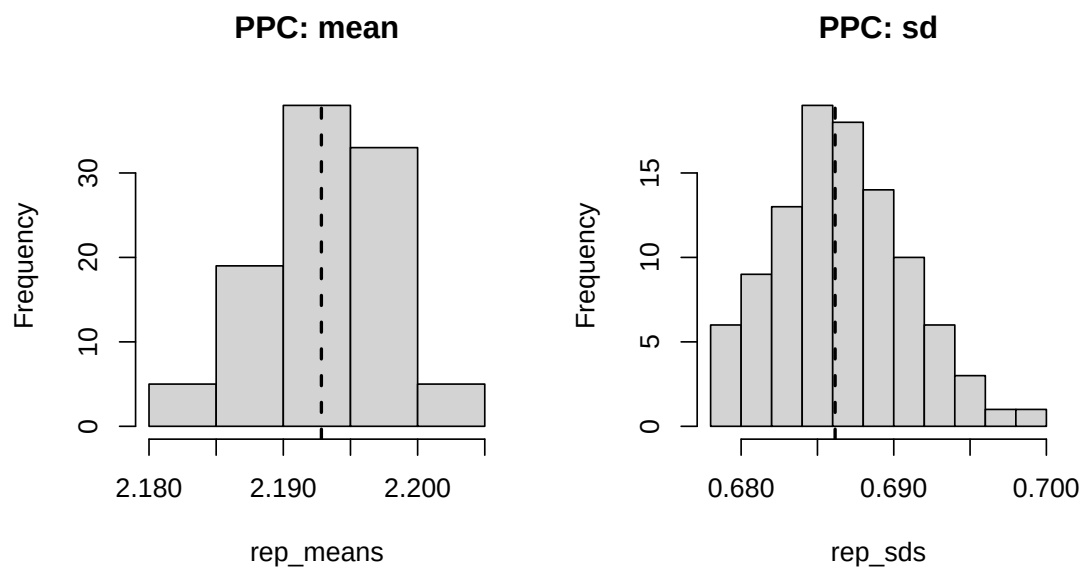


Figure 7: Posterior predictive checks based on summary statistics. The observed mean and standard deviation fall well within the distribution of replicated values.

Overall, the hierarchical Bayesian framework provides a flexible and effective approach for modeling air pollution data. Possible extensions include the incorporation of spatial dependence across stations, the use of more flexible likelihood specifications, and the inclusion of additional covariates capturing environmental and meteorological factors.

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